University of Windsor

Department of Electrical and Computer Engineering

GENG-8030 - MATLAB

Tutorial Questions and MATLAB Solutions

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Tutorial 02: Numeric, Cell and Structure Arrays

Questions in this tutorial

- Problem 27: springs, compression, and stored energy
- Problem 41: solve a linear system by left division

Problem 27

The potential energy stored in a spring is $kx^2/2$, where k is the spring constant and x is the compression in the spring. The force required to compress the spring is kx. The following table gives the data for five springs. Use MATLAB to find:

a. the compression x in each spring;

b. the potential energy stored in each spring.

Spring	1	2	3	4	5
Force (N)	11	7	8	10	9
Spring constant k (N/m)	1000	600	900	1300	700

MATLAB Solution

```
clear; clc;

F = [11 7 8 10 9];            % N
k = [1000 600 900 1300 700];  % N/m

% Compression from F = kx
x = F ./ k;

% Potential energy in each spring
PE = 0.5 .* k .* x.^2;

results = table((1:5).', F.', k.', x.', PE.', ...
    'VariableNames', {'Spring','Force_N','k_N_per_m','Compression_m','PE_J'});
disp(results);
```

Problem 41

Solve the following problem using the left-division method:

$$
\begin{aligned}
6x - 3y + 4z &= 41, \\
12x + 5y - 7z &= -26, \\
-5x + 2y - 6z &= 16.
\end{aligned}
$$

MATLAB Solution

```
clear; clc;

A = [ 6  -3   4;
     12   5  -7;
     -5   2  -6];

b = [41; -26; 16];

solution = A \ b;

fprintf('x = %.6f\n', solution(1));
fprintf('y = %.6f\n', solution(2));
fprintf('z = %.6f\n', solution(3));
```

Tutorial 03: Functions

Questions in this tutorial

- Problem 17: conical paper cup area minimization
- Problem 25: anonymous function and minimum search

Problem 17

The volume V and paper surface area A of a conical paper cup are given by

$$V = \frac{1}{3}\pi r^2 h, \qquad A = \pi r\sqrt{r^2 + h^2},$$

where r is the radius of the base of the cone and h is the height of the cone.

a. By eliminating h, obtain the expression for A as a function of r and V.

b. Create a user-defined function that accepts r as the only argument and computes A for a given value of V. Declare V to be global within the function.

c. For $V = 10\ \text{in}^3$, use the function with `fminbnd` to compute the value of r that minimizes the area A. What is the corresponding value of the height h? Investigate the sensitivity of the solution by plotting A versus r. How much can r vary about its optimal value before the area increases 10% above its minimum value?

MATLAB Solution

```
clear; clc; close all;
global V

V = 10;  % in^3

% A(r) after eliminating h:
% h = 3V/(pi*r^2)
% A(r) = pi*r*sqrt(r^2 + (3V/(pi*r^2))^2)

[r_opt, A_min] = fminbnd(@cone_area, 0.05, 10);
h_opt = 3 * V / (pi * r_opt^2);

r = linspace(0.2, 3.0, 500);
A = cone_area(r);
target = 1.10 * A_min;

idx = find(A ≤ target);
r_low = r(idx(1));
r_high = r(idx(end));

fprintf('Optimal radius      = %.6f in\n', r_opt);
fprintf('Corresponding height= %.6f in\n', h_opt);
fprintf('Minimum area        = %.6f in^2\n', A_min);
fprintf('10%% area band       = [%.6f, %.6f] in\n', r_low, r_high);

plot(r, A, 'LineWidth', 1.5); hold on;
```

```
plot(r_opt, A_min, 'ro', 'MarkerFaceColor', 'r');
yline(target, '--k', '110% of minimum');
grid on;
xlabel('r (in)');
ylabel('A(r) (in^2)');
title('Cone Surface Area vs Radius');

function A = cone_area(r)
    global V
    h = 3 * V ./ (pi .* r.^2);
    A = pi .* r .* sqrt(r.^2 + h.^2);
end
```

Problem 25

Create an anonymous function for $30x^2 - 300x + 4$ and use it:

a. to plot the function and determine the approximate location of its minimum;

b. with the `fminbnd` function to precisely determine the location of the minimum.

MATLAB Solution

```
clear; clc; close all;

f = @(x) 30*x.^2 - 300*x + 4;

x = linspace(-2, 12, 400);
y = f(x);

[x_min, y_min] = fminbnd(f, -2, 12);

plot(x, y, 'LineWidth', 1.5); hold on;
plot(x_min, y_min, 'ro', 'MarkerFaceColor', 'r');
grid on;
xlabel('x');
ylabel('f(x)');
title('f(x) = 30x^2 - 300x + 4');

fprintf('Minimum occurs at x = %.6f\n', x_min);
fprintf('Minimum value is %.6f\n', y_min);
```

Tutorial 04: Programming with MATLAB

Questions in this tutorial

- Problem 9: compare two stock-price arrays
- Problem 40: cable tensions and feasible cable length

Problem 9

The arrays `price_A` and `price_B` given below contain the price in dollars of two stocks over 10 days. Use MATLAB to determine how many days the price of stock A was below the price of stock B.

```
price_A = [19, 18, 22, 21, 25, 19, 17, 21, 27, 29]

price_B = [22, 17, 20, 23, 24, 18, 16, 25, 28, 27]
```

MATLAB Solution

```
clear; clc;

price_A = [19, 18, 22, 21, 25, 19, 17, 21, 27, 29];
price_B = [22, 17, 20, 23, 24, 18, 16, 25, 28, 27];

days_below = price_A < price_B;
num_days = sum(days_below);

fprintf('Days A was below B = %d\n', num_days);
```

Problem 40

A weight W is supported by two cables anchored a distance D apart. The cable length L_{AB} is given, but the length L_{AC} is to be selected. Each cable can support a maximum tension force equal to W. For the weight to remain stationary, the total horizontal force and total vertical force must each be zero. This principle gives the equations

$$-T_{AB}\cos\theta + T_{AC}\cos\phi = 0, \qquad T_{AB}\sin\theta + T_{AC}\sin\phi = W.$$

We can solve these equations for the tension forces T_{AB} and T_{AC} if we know the angles θ and ϕ. From the law of cosines,

$$\theta = \cos^{-1}\left(\frac{D^2 + L_{AB}^2 - L_{AC}^2}{2DL_{AB}}\right).$$

From the law of sines,

$$\phi = \sin^{-1}\left(\frac{L_{AB}\sin\theta}{L_{AC}}\right).$$

For the given values $D = 6$ ft, $L_{AB} = 3$ ft, and $W = 2000$ lb, use a `while` loop in MATLAB to find $L_{AC,\min}$, the shortest length L_{AC} that can be used without T_{AB} or T_{AC} exceeding 2000 lb. Plot the tension forces T_{AB} and T_{AC} on the same graph versus L_{AC} for $L_{AC,\min} < L_{AC} < 6.7$.

MATLAB Solution

```matlab
clear; clc; close all;

D = 6;           % ft
LAB = 3;         % ft
W = 2000;        % lb
LAC_max = 6.7;   % ft
dL = 1e-3;

LAC = abs(D - LAB) + 1e-4;
TAB = inf;
TAC = inf;

while (TAB > W) || (TAC > W)
    theta = acos((D^2 + LAB^2 - LAC^2) / (2 * D * LAB));
    phi = asin((LAB * sin(theta)) / LAC);

    A = [-cos(theta),  cos(phi);
          sin(theta),  sin(phi)];
    b = [0; W];

    T = A \ b;
    TAB = T(1);
    TAC = T(2);

    if (TAB > W) || (TAC > W)
        LAC = LAC + dL;
    end

    if LAC > LAC_max
        error('No feasible LAC found up to LAC_max.');
    end
end

LAC_min = LAC;
fprintf('LAC_min = %.4f ft\n', LAC_min);
fprintf('TAB = %.2f lb, TAC = %.2f lb\n', TAB, TAC);

Lvec = LAC_min:dL:LAC_max;
TABv = zeros(size(Lvec));
TACv = zeros(size(Lvec));

for k = 1:length(Lvec)
    LAC = Lvec(k);
    theta = acos((D^2 + LAB^2 - LAC^2) / (2 * D * LAB));
    phi = asin((LAB * sin(theta)) / LAC);

    A = [-cos(theta),  cos(phi);
          sin(theta),  sin(phi)];
    T = A \ [0; W];

    TABv(k) = T(1);
    TACv(k) = T(2);
end

plot(Lvec, TABv, 'LineWidth', 1.5); hold on;
plot(Lvec, TACv, 'LineWidth', 1.5);
```

```
yline(W, '--k', '2000 lb limit');
grid on;
xlabel('L_{AC} (ft)');
ylabel('Tension (lb)');
legend('T_{AB}', 'T_{AC}', 'Location', 'best');
title('Cable Tensions vs L_{AC}');
```

Tutorial 05: Advanced Plotting

Questions in this tutorial

- Problem 19*: projectile height and trajectory plots
- Problem 42: surface and contour plots of a temperature field

Problem 19*

The height $h(t)$ and horizontal distance $x(t)$ traveled by a ball thrown at an angle A with a speed v are given by

$$h(t) = vt \sin A - \frac{1}{2} g t^2, \qquad x(t) = vt \cos A.$$

At Earth's surface the acceleration due to gravity is $g = 9.81$ m/s^2.

a. Suppose the ball is thrown with a velocity $v = 20$ m/s at an angle of 25°. Use MATLAB to compute how high the ball will go, how far it will go, and how long it will take to hit the ground.

b. Use the values of v and A given in part (a) to plot the ball's trajectory; that is, plot h versus x for positive values of t.

c. Plot the trajectories for $A = 45°$ corresponding to five values of the initial velocity v: 20, 24, 28, 32, and 36 m/s.

MATLAB Solution

```
clear; clc; close all;

g = 9.81;
v = 20;
A = deg2rad(25);

t_flight = 2 * v * sin(A) / g;
h_max = (v * sin(A))^2 / (2 * g);
range = v^2 * sin(2 * A) / g;

fprintf('Maximum height = %.6f m\n', h_max);
fprintf('Range          = %.6f m\n', range);
fprintf('Flight time    = %.6f s\n', t_flight);

t = linspace(0, t_flight, 300);
h = v * t * sin(A) - 0.5 * g * t.^2;
x = v * t * cos(A);

figure;
plot(x, h, 'LineWidth', 1.5);
grid on;
xlabel('x (m)');
ylabel('h (m)');
title('Trajectory for v = 20 m/s, A = 25 deg');
```

```

figure;
velocities = [20 24 28 32 36];
A2 = deg2rad(45);

for k = 1:length(velocities)
    v0 = velocities(k);
    tf = 2 * v0 * sin(A2) / g;
    t = linspace(0, tf, 300);
    h = v0 * t * sin(A2) - 0.5 * g * t.^2;
    x = v0 * t * cos(A2);
    plot(x, h, 'LineWidth', 1.5); hold on;
end

grid on;
xlabel('x (m)');
ylabel('h (m)');
legend('20','24','28','32','36', 'Location', 'best');
title('Trajectories for A = 45 deg');
```

Problem 42

A square metal plate is heated to 80°C at the corner corresponding to $x = y = 1$. The temperature distribution in the plate is described by

$$T = 80e^{-(x-1)^2-3(y-1)^2}.$$

Obtain the surface and contour plots for the temperature. Label each axis. What is the temperature at the corner corresponding to $x = y = 0$?

MATLAB Solution

```
clear; clc; close all;

x = linspace(0, 1, 101);
y = linspace(0, 1, 101);
[X, Y] = meshgrid(x, y);

T = 80 * exp(-(X - 1).^2 - 3 * (Y - 1).^2);
T00 = 80 * exp(-(0 - 1)^2 - 3 * (0 - 1)^2);

fprintf('Temperature at x = 0, y = 0 is %.6f degC\n', T00);

figure;
surf(X, Y, T);
shading interp;
colorbar;
xlabel('x');
ylabel('y');
zlabel('T (degC)');
title('Surface Plot of Temperature');

figure;
contourf(X, Y, T, 20);
colorbar;
```

```
xlabel('x');
ylabel('y');
title('Contour Plot of Temperature');
```

Tutorial 06: Model Building and Regression

Questions in this tutorial

- Problem 7: capacitor-voltage regression from data
- Problem 9: polynomial curve fits for drying time

Problem 7

A certain electric circuit has a resistor and a capacitor. The capacitor is initially charged to 100 V. When the power supply is detached, the capacitor voltage decays with time, as the following data table shows. Find a functional description of the capacitor voltage v as a function of time t. Plot the function and the data on the same plot.

Time (s)	0	0.5	1	1.5	2	2.5	3	3.5	4
Voltage (V)	100	62	38	21	13	7	4	2	3

MATLAB Solution

```
clear; clc; close all;

t = [0 0.5 1 1.5 2 2.5 3 3.5 4];
v = [100 62 38 21 13 7 4 2 3];

% Exponential model v(t) = a*exp(b*t)
p = polyfit(t, log(v), 1);
b = p(1);
a = exp(p(2));

t_fine = linspace(min(t), max(t), 400);
v_fit = a * exp(b * t_fine);
v_data_fit = a * exp(b * t);

fprintf('Best-fit model: v(t) = %.6f * exp(%.6f * t)\n', a, b);

plot(t, v, 'ko', 'MarkerFaceColor', 'k'); hold on;
plot(t_fine, v_fit, 'b-', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Voltage (V)');
legend('Data', 'Exponential fit', 'Location', 'best');
title('Capacitor Voltage Decay');
```

Problem 9

The following data give the drying time T of a certain paint as a function of the amount of a certain additive A.

a. Find the first-, second-, third-, and fourth-degree polynomials that fit the data, and plot each polynomial with the data. Determine the quality of the curve fit for each by computing J, S, and r.

b. Use the polynomial giving the best fit to estimate the amount of additive that minimizes the drying time.

A (oz)	0	1	2	3	4	5	6	7	8	9
T (min)	130	115	110	90	89	89	95	100	110	125

MATLAB Solution

```
clear; clc; close all;

A = 0:9;
T = [130 115 110 90 89 89 95 100 110 125];
n = numel(A);
SS_tot = sum((T - mean(T)).^2);

figure;
plot(A, T, 'ko', 'MarkerFaceColor', 'k'); hold on;

degrees = 1:4;
J = zeros(size(degrees));
S = zeros(size(degrees));
r = zeros(size(degrees));
coeffs = cell(size(degrees));

x_fine = linspace(min(A), max(A), 400);

for k = 1:length(degrees)
    deg = degrees(k);
    coeffs{k} = polyfit(A, T, deg);
    T_hat = polyval(coeffs{k}, A);
    T_fine = polyval(coeffs{k}, x_fine);

    J(k) = sum((T - T_hat).^2);
    S(k) = sqrt(J(k) / (n - (deg + 1)));
    r(k) = sqrt(max(0, 1 - J(k) / SS_tot));

    plot(x_fine, T_fine, 'LineWidth', 1.2);
end

grid on;
xlabel('Additive A (oz)');
ylabel('Drying time T (min)');
legend('Data', '1st', '2nd', '3rd', '4th', 'Location', 'best');
title('Polynomial Fits to Drying-Time Data');

disp(table(degrees.', J.', S.', r.', ...
    'VariableNames', {'Degree','J','S','r'}));

[¬, best_idx] = max(r);
best_deg = degrees(best_idx);
best_poly = coeffs{best_idx};
best_fun = @(x) polyval(best_poly, x);

[A_opt, T_min] = fminbnd(best_fun, min(A), max(A));
```

```

fprintf('Best polynomial degree = %d\n', best_deg);
fprintf('Minimum drying time occurs near A = %.6f oz\n', A_opt);
fprintf('Estimated minimum drying time = %.6f min\n', T_min);
```

Tutorial 07: Interpolation

Questions in this tutorial

- Problem 27: linear interpolation with missing temperatures

Problem 27

The following table gives temperature data in °C as a function of time of day and day of the week at a specific location. Data are missing for the entries marked with a question mark (?). Use linear interpolation with MATLAB to estimate the temperature at the missing points.

Hour	Mon	Tues	Wed	Thurs	Fri
1	16	15	12	17	16
2	13	?	8	11	12
3	14	15	9	?	15
4	17	15	14	17	19
5	21	18	19	20	24

MATLAB Solution

```
clear; clc;

hours = 1:5;

tues = [15 NaN 15 15 18];
thurs = [17 11 NaN 17 20];

tues_missing = interp1(hours(¬isnan(tues)), tues(¬isnan(tues)), 2, 'linear');
thurs_missing = interp1(hours(¬isnan(thurs)), thurs(¬isnan(thurs)), 3, 'linear');

fprintf('Estimated Tuesday temperature at hour 2 = %.6f degC\n', tues_missing);
fprintf('Estimated Thursday temperature at hour 3 = %.6f degC\n', thurs_missing);
```

Tutorial 08: Linear Systems

Questions in this tutorial

- Problem 16: traffic-flow solvability and sensor placement
- Problem 22: interpolate quadratic and cubic polynomials

Problem 16

See Figure P16. Assume that no vehicles stop within the network. A traffic engineer wants to know if the traffic flows can be computed given the measured flows shown in the figure. If not, determine how many more traffic sensors need to be installed, and obtain the expressions for the other traffic flows in terms of the measured quantities.

MATLAB Solution

```
clear; clc;

% Unknowns are x = [f1 f2 f3 f4 f5 f6 f7]^T
% Flow-balance equations from the six intersections:
% Node 1: f1 + f3 = 300
% Node 2: f1 - f2 - f4 = -300
% Node 3: f2 + f5 = 600
% Node 4: f3 + f6 = 400
% Node 5: f4 - f6 + f7 = 200
% Node 6: f5 + f7 = 600

A = [1  0  1  0  0  0  0;
     1 -1  0 -1  0  0  0;
     0  1  0  0  1  0  0;
     0  0  1  0  0  1  0;
     0  0  0  1  0 -1  1;
     0  0  0  0  1  0  1];

b = [300; -300; 600; 400; 200; 600];

rankA = rank(A);
num_unknowns = size(A, 2);
extra_sensors = num_unknowns - rankA;

fprintf('Rank of coefficient matrix = %d\n', rankA);
fprintf('Unknown traffic flows      = %d\n', num_unknowns);
fprintf('Additional sensors needed  = %d\n', extra_sensors);

% Let the additional sensors measure f3 = s and f4 = t.
% Then every other flow can be written in terms of s and t.
syms s t
f3 = s;
f4 = t;
f1 = 300 - s;
f2 = 600 - s - t;
f5 = s + t;
f6 = 400 - s;
f7 = 600 - s - t;
```

```

disp('General traffic-flow solution:')
disp(table(f1, f2, f3, f4, f5, f6, f7))
```

Problem 22

a. Use MATLAB to find the coefficients of the quadratic polynomial $y = ax^2 + bx + c$ that passes through the three points $(x, y) = (1, 4), (3, 73), (5, 120)$.

b. Use MATLAB to find the coefficients of the cubic polynomial $y = ax^3 + bx^2 + cx + d$ that passes through the three points given in part (a).

MATLAB Solution

```
clear; clc;

x = [1; 3; 5];
y = [4; 73; 120];

% Part (a): unique quadratic
A_quad = [x.^2, x, ones(size(x))];
quad_coeffs = A_quad \ y;

fprintf('Quadratic coefficients [a b c]^T:\n');
disp(quad_coeffs);

% Part (b): cubic is underdetermined with only three points.
% Solve symbolically for a family of cubics in terms of free parameter d.
syms a b c d
eqns = [a*x(1)^3 + b*x(1)^2 + c*x(1) + d == y(1);
        a*x(2)^3 + b*x(2)^2 + c*x(2) + d == y(2);
        a*x(3)^3 + b*x(3)^2 + c*x(3) + d == y(3)];

sol = solve(eqns, [a b c]);

fprintf('General cubic coefficients:\n');
fprintf('a = %s\n', char(sol.a));
fprintf('b = %s\n', char(sol.b));
fprintf('c = %s\n', char(sol.c));
fprintf('d is free\n');

% One particular cubic solution: choose d = 0
particular = double([subs(sol.a, d, 0);
                     subs(sol.b, d, 0);
                     subs(sol.c, d, 0);
                     0]);

fprintf('One particular cubic [a b c d]^T with d = 0:\n');
disp(particular);
```

Tutorial 09: Numerical Methods

Questions in this tutorial

- Problem 44: initial and step responses from a state model
- Problem 45: free, step, and total response of a second-order system

Problem 44

The following state model describes the motion of a certain mass connected to a spring, with viscous friction on the surface, where $m = 1$, $c = 2$, and $k = 5$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f(t).$$

a. Use the `initial` function to plot the position x_1 of the mass if the initial position is 5 and the initial velocity is 3.

b. Use the `step` function to plot the step response of the position and velocity for zero initial conditions, where the magnitude of the step input is 10.

MATLAB Solution

```
clear; clc; close all;

A = [0 1; -5 -2];
B = [0; 1];
C = eye(2);
D = [0; 0];

sys = ss(A, B, C, D);

figure;
subplot(2,1,1);
initial(sys(1), [5; 3]);
grid on;
title('Initial Response of Position x_1');

subplot(2,1,2);
step(10 * sys);
grid on;
title('Step Response of Position and Velocity (Input = 10)');
```

Problem 45

Consider the following equation:

$$5\ddot{y} + 2\dot{y} + 10y = f(t).$$

a. Plot the free response for the initial conditions $y(0) = 10$ and $\dot{y}(0) = -5$.

b. Plot the unit-step response for zero initial conditions.

c. The total response to a step input is the sum of the free response and the step response. Demonstrate this fact for this equation by plotting the sum of the solutions found in parts (a) and (b) and comparing the plot with that generated by solving for the total response with $y(0) = 10$ and $\dot{y}(0) = -5$.

MATLAB Solution

```
clear; clc; close all;

ode_free = @(t, x) [x(2); (-2*x(2) - 10*x(1)) / 5];
ode_step = @(t, x) [x(2); (1 - 2*x(2) - 10*x(1)) / 5];

tspan = [0 15];

[t_free, x_free] = ode45(ode_free, tspan, [10; -5]);
[t_step, x_step] = ode45(ode_step, tspan, [0; 0]);
[t_total, x_total] = ode45(ode_step, tspan, [10; -5]);

y_sum = x_free(:,1) + x_step(:,1);

plot(t_free, x_free(:,1), '--', 'LineWidth', 1.4); hold on;
plot(t_step, x_step(:,1), ':', 'LineWidth', 1.4);
plot(t_total, x_total(:,1), 'k-', 'LineWidth', 1.8);
plot(t_free, y_sum, 'r-.', 'LineWidth', 1.4);
grid on;
xlabel('Time');
ylabel('y(t)');
legend('Free response', 'Step response', 'Total response', 'Free + step', ...
    'Location', 'best');
title('Verification of Superposition');
```

Tutorial 10: Simulink

Questions in this tutorial

- Problem 8: second-order ODE with sinusoidal forcing
- Problem 15: coupled first-order system with sinusoidal input
- Problem 13: coupled first-order system with ramp input

Problem 8

Create a Simulink model to plot the solution of the following equation for $0 < t < 6$:

$$10\ddot{y} = 7\sin(4t) + 5\cos(3t), \qquad y(0) = 3, \qquad \dot{y}(0) = 2.$$

MATLAB Solution

```
clear; clc; close all;

odefun = @(t, x) [x(2);
                  (7*sin(4*t) + 5*cos(3*t)) / 10];

[t, x] = ode45(odefun, [0 6], [3; 2]);

plot(t, x(:,1), 'LineWidth', 1.5);
grid on;
xlabel('t');
ylabel('y(t)');
title('Solution of 10y'''' = 7sin(4t) + 5cos(3t)');
```

Problem 15

Construct a Simulink model to plot the solution of the following equations for $0 < t < 10$:

$$\dot{x} = -5x + 3y + 5\sin(2t), \qquad x(0) = 0,$$
$$\dot{y} = 3x - 4y, \qquad y(0) = 0.$$

MATLAB Solution

```
clear; clc; close all;

odefun = @(t, z) [-5*z(1) + 3*z(2) + 5*sin(2*t);
                   3*z(1) - 4*z(2)];

[t, z] = ode45(odefun, [0 10], [0; 0]);

plot(t, z(:,1), 'LineWidth', 1.5); hold on;
plot(t, z(:,2), 'LineWidth', 1.5);
grid on;
xlabel('t');
ylabel('States');
legend('x(t)', 'y(t)', 'Location', 'best');
title('Solution of the Coupled First-Order System');
```

Problem 13

Construct a Simulink model to plot the solution of the following equations for $0 < t < 2$:

$$\dot{x}_1 = -6x_1 + 4x_2,$$

$$\dot{x}_2 = 5x_1 - 7x_2 + f(t), \qquad \text{where } f(t) = 3t.$$

Use the Ramp block in the Sources library.

MATLAB Solution

```
clear; clc; close all;

odefun = @(t, x) [-6*x(1) + 4*x(2);
                   5*x(1) - 7*x(2) + 3*t];

[t, x] = ode45(odefun, [0 2], [0; 0]);

plot(t, x(:,1), 'LineWidth', 1.5); hold on;
plot(t, x(:,2), 'LineWidth', 1.5);
grid on;
xlabel('t');
ylabel('States');
legend('x_1(t)', 'x_2(t)', 'Location', 'best');
title('Solution with Ramp Input f(t) = 3t');
```